Understanding Algae Growth Problem in Kansas Farm Ponds

Introduction

This document explains the process of restoring a small farm pond that had become covered with algae. The pond was last rebuilt in 1963 and had accumulated more than sixty years of silt, fertilizer runoff, and organic material. Shallow water and warmer weather patterns contributed to rapid algae growth, making the pond unsightly and less functional. This project documents the dredging process, explains why dredging was the most effective long-term solution, and shows the results using original photographs. The goal is to help new rural property owners understand why ponds turn green and what is involved in restoring them.

Section 1: Why the Pond Turned Green

The pond turned green because several long-term environmental factors combined to create ideal conditions for algae growth. Over more than sixty years, fertilizer runoff from surrounding fields added nutrients that feed algae. Warmer weather patterns have also increased water temperatures, which accelerates algae reproduction. The most important factor, however, was the buildup of silt that made the pond increasingly shallow. As shown in Table 1, shallow water allows more sunlight to reach the bottom, which significantly increases algae growth. Of the three causes, dredging to restore depth was the only factor I could directly control.

Table 1. Causes of Algae Growth in Kansas Ponds.

Cause	How It Contributes	Can the Owner Control It?
Fertilizer runoff	Adds nutrients (nitrogen & phosphorus that feed algae growth	Partially- depends on surrounding land use
Climate warming	Warmer water increases algae reproduction rates	No (Not by him/herself)
Shallow depth from silt buildup	More sunlight reaches the pond bottom, accelerating algae growth	Yes- by dredging (rebuilding)

Section 2: Why Dredging Was the Best Solution

Several methods can be used to control algae in small ponds, but most of them are temporary and require repeated maintenance. Chemical treatments can clear the water quickly, but the algae usually return once the chemicals wear off. Aeration improves water quality but does not remove the silt that causes shallow conditions. Barley straw and skimming offer only minor or short-term improvements. As shown in Table 2, dredging was the only method that addressed the root cause of the problem by removing decades of accumulated silt and restoring the pond's depth. Although dredging is the most expensive option, it provides the most effective and long-lasting solution.

Table 2. Comparison of Algae Control Methods.

Method	Cost	Effectiveness	Temporary or Long-Term	Notes
Chemical treatments	Moderate	High at first, but algae returns	Temporary	Requires repeated applications
Aeration	Moderate to high	Helps long-term water quality	Long-Term	Needs electricity and maintenance
Barley straw	Low	Mild effectiveness	Temporary	Works slowly; not reliable for heavy blooms
Skimming	Very low	Removes surface algae only	Temporary	Labor-intensive; doesn't fix root causes
Dredging (my method)	High	Very high	Long-Term	Removes silt, deepens pond, reduces sunlight penetration

Section 3: The Pond Clean-Out Process

The following photos document the major stages of the pond clean-out process. Over several days, the bulldozer removed decades of accumulated silt, reshaped the pond edges, and restored the original clay bottom. These images show the progression from the algae-covered pond to the fully restored basin and finally to the refilled pond. Together, they illustrate how dredging directly addresses the root causes of algae growth by increasing depth and improving water quality.

Figure 1. Pond before dredging. (Photo by author.)

Figure 2. Cutting the bank. (Photo by author.)

Figure 3. During excavation. (Photo by author.)

Figure 4. After excavation. (Photo by author.)

Figure 5. Full pond. (Photo by author.)

Section 4: Estimated Increase in Pond Volume

Although exact measurements were not taken before the project, reasonable estimates based on the pond's dimensions and standard farm-pond volume formulas show a significant increase in water capacity after dredging. As shown in Table 3, the pond held an estimated 3,768 cubic feet of water before dredging and approximately 13,846 cubic feet afterward. This represents an increase of more than 10,000 cubic feet, or roughly 75,000 gallons of additional water. The deeper basin reduces sunlight penetration and improves long-term water quality, making the pond far less susceptible to algae growth.

Table 3. Estimated Pond Volume Before and After Dredging.

Condition	Estimated Avg Depth	Estimated Volume (cubic ft)	Estimated Volume (gallons)	Notes
Before dredging	2.5 ft	3,768 cu ft	~28,000 gallons	Shallow, algae-prone
After dredging	7 ft	13,846 cu ft	~103,000 gallons	Restored depth, improved water quality
Increase	—	**+10,078 cu ft**	**+75,000 gallons**	Much larger water capacity

Conclusion

Restoring the pond required addressing the long-term environmental factors that caused excessive algae growth. Fertilizer runoff, warmer weather patterns, and decades of silt buildup created shallow conditions that allowed algae to thrive. Dredging was the most effective solution because it removed the accumulated silt, restored the pond's depth, and increased its overall water capacity. The photos and volume estimates demonstrate how the project improved water quality and reduced the likelihood of future algae problems. This restoration provides a long-term solution that will keep the pond healthier and more functional for years to come.

Works Cited

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